Paediatric acute liver failure: a multidisciplinary perspective on when a critically ill child is unsuitable for liver transplantation

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Paediatric acute liver failure is a devastating condition with high morbidity and mortality, which is challenging to manage for the hepatologist, intensivist, and associated specialists. Emergency liver transplantation is required for 10–20% of patients, but for 10% of critically ill children, liver transplantation is deemed unsuitable; the child might be too unwell, or the underlying cause might carry a poor prognosis. Other social, logistical, or ethical considerations are often relevant. Liver transplantation when a patient is too unwell creates perioperative risk to the child that could lead to morbidity, mortality, and potential graft wastage, which is detrimental for others on the waiting list. Donor liver scarcity should prompt an evaluation of whether a transplant is justified through a holistic multidisciplinary lens that considers medical, social, logistical, and ethical concerns. In this Review, we explore, from a multidisciplinary perspective, why a critically unwell child with paediatric acute liver failure might be unsuitable for liver transplantation.

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Introduction

Paediatric acute liver failure (PALF) is associated with high morbidity and mortality, ranging from 10% to 40% depending on the clinical and geographical setting.[1,2] The condition is defined by biochemical liver injury in a child with no medical history of liver disease but presenting with high international normalised ratio ([INR] >2·0 [normal range 0·8–1·1); alternatively, INR >1·5 in the presence of hepatic encephalopathy; unresponsive to vitamin K).[2] According to a Pediatric Health Information System database study, the proportion of patients who survive PALF with native liver has improved from 15% in the 1980s, to 73% in 2008–13,[3–5] which reflects advances in multisystem supportive critical care management such as extracorporeal therapies and neuroprotective measures.[6,7] In most cases, it is now reasonable and realistic to await organ recovery, rather than automatically proceeding to liver transplantation. Consequently, most children with PALF are now not listed for liver transplant. However, many clinically unstable patients either require liver transplantation (patients with PALF comprise 10% of all paediatric liver transplantations in the USA[8]) or are too critically unwell to receive one. These patients are the focus of this Review.

According to the Pediatric Acute Liver Failure Study Group (PALFSG) in the USA, 29% of patients (44/151) in the national PALF registry between 2012–14 were listed for a transplant, and 17% (26/151) received a transplant within 7 days of registry enrolment.[1] However, between 1999 and 2018, 8% of patients (95/1144) were deemed too sick for transplantation. Although this patient category was not consistently defined and probably included patients who were medically unsuitable alongside those with social or logistical contraindications,[1] 7-day mortality between 2008 and 2014 was 29% (23/79), and 30-day mortality exceeded 60% (50/79).[1]

Transplant candidates must be carefully selected, not least because of the long waitlist for donor livers and the risks of liver transplantation, both acutely and life-long, owing to immunosuppression. Transplant assessments identify patients who will die without a liver transplant and who have a good chance of surviving with satisfactory quality of life with liver transplant. In the UK, the National Health Service Blood and Transplant states that patients are eligible for a liver transplant if their probability of 5-year post-transplant survival with an acceptable quality of life is greater than 50%.[9] Patients with multiorgan failure secondary to PALF might not meet this threshold. Non-clinical factors, such as social, psychological, logistical, and ethical considerations, also affect listing decisions.

Patients for whom transplantation cannot be performed at the time of assessment because of clinical instability typically have three options: (1) continued intensive care as a bridge to transplant eligibility or potential recovery; (2) emergency hepatectomy, if there is the potential for hepatectomy to lead to stabilisation

Key messages

- Paediatric acute liver failure (PALF) is associated with high morbidity and mortality, ranging from 10% to 40% depending on the clinical and geographical setting; 10–20% of patients will need a liver transplant
- Transplant candidates must be carefully selected; a critically ill child's suitability for transplantation should be based on a multidisciplinary team's perspectives on the cause of PALF, multiorgan dysfunction, surgical difficulties, and physiological stability to survive transplantation
- Formal ethics reviews can support difficult clinical decision making and moral choices in relation to liver transplantation for a patient with PALF, and must consider the patient's quantity and quality of life; an ethical framework can also guide equitable practice, especially in settings where organs for transplantation are in shortage

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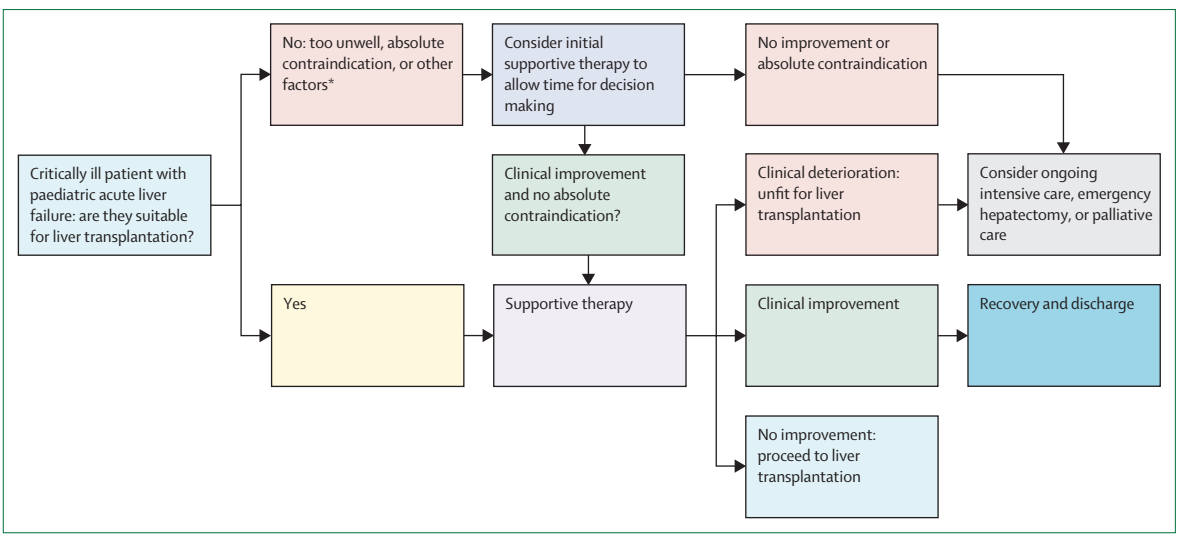

Figure 1: **Assessing the suitability of liver transplantation for a patient with paediatric acute liver failure**
Figure created with BioRender.com. *Other factors include social, psychological, and economic considerations.

and transplant eligibility; or (3) palliative and end-of-life care (figure 1).[10,11] Decision making is complex and often stressful, especially when transplantation is considered unsuitable. Best practice is therefore for a large, experienced, multidisciplinary team to work alongside the patient's family. The team should ideally include hepatologists, intensivists, anaesthetists, transplant surgeons, specialist nurses, family liaison professionals, social workers, and clinical ethicists. Although each professional brings a unique perspective to the child's care, the team as a whole must collaborate to consider key aspects of PALF treatment strategy: PALF aetiology and prognosis, safety and efficacy of liver transplantation with careful consideration of intensive care, anaesthesia, and surgery needs, risk of perioperative death, the patient and their family's social and psychological circumstances, and ethical dimensions of providing care in the child's best interest.

Hepatology considerations: aetiology, pathogenesis, and prognosis

Hepatologists treating patients with PALF are often responsible for advising on optimal use of a precious resource—the donated organ—while minimising patient suffering. Although guidelines can aid evaluation of children for liver transplantation,[12] the decision to list a patient for a transplant ultimately resides with individual centres and requires not only alignment between physician experience and clinical assessment but organ availability. Unfortunately, given the scarcity of organs, not every patient who might benefit from a liver transplant can receive one.

Aetiology

A child with PALF raises several pressing clinical questions for the managing team (figure 2; panel 1). The aetiology is often the hepatologist's first consideration in the overall assessment of the patient. If a cause is identified, what is the prognosis? If prognosis is poor, is a liver transplant necessary for survival, or might liver transplantation even change the prognosis? Ultimately, the degree of acuity (or sickness) might not affect transplant listing decisions as much as aetiology, as each aetiology has a different prognosis.

Aetiology varies by age, and generally, the younger the child with PALF (especially neonates and infants), the higher the mortality risk.[13] Paracetamol overdose is associated with a high chance of survival with native liver, even for patients with severe hepatic encephalopathy.[14] By contrast, chances of survival are low with haematological or oncological causes, such as veno-occlusive disease or haemophagocytic lymphohistiocytosis. Generalised extrahepatic malignancy, multisystemic mitochondrial disease, and uncontrolled systemic infection (especially fungal or viral) are considered absolute contraindications to liver transplantation (panel 2).[15]

Virus-induced PALF has variable outcomes. Oncology patients with herpes virus infections (eg, varicella zoster virus or herpes simplex virus) have very poor prognoses, whereas outcomes tend to be more favourable with hepatitis A virus infection.[16,17] Some aetiologies might have modifiable prognoses if aetiology-specific treatment can be commenced, such as copper chelation in Wilson's disease or steroids in $CD8^+$ cytotoxic T-cell predominant liver failure.[18]

In cases where PALF develops secondary to a multiorgan insult leading to hypoxic or ischaemic hepatitis (as seen in dengue and septic shock), awaiting resolution of the underlying disease or trigger can be necessary before considering listing. Finally, the absence of a known diagnosis could precipitate the need for

Figure 2: **Multidisciplinary considerations in managing a child with paediatric acute liver failure in deteriorating condition**
Figure created with BioRender.com. GRWR=graft–recipient weight ratio.

liver transplantation because indeterminate PALF is associated with poor outcomes.[1,19]

Prognostication

For the hepatologist, establishing when a child is unsuitable for liver transplantation remains challenging without tools to predict the dynamic disease course. Prognostic modelling to aid in PALF disease severity assessment and risk estimation is therefore a key research focus. Combinations of biochemical tests (eg, lactate, total bilirubin, phosphate, INR, prothrombin time, ammonia, vitamin D-binding protein), clinical features (eg, encephalopathy, cerebral oedema, age, growth), and diagnosis (eg, paracetamol, Wilson's disease, hepatitis A virus) have been evaluated with mixed success.[15,16,20–23] Most data indicate that high total bilirubin count, INR, and high pretransplantation lactate (>5 mmol/L) are risk factors for early mortality after liver transplantation.

Unfortunately, the predictive utility of PALF models that incorporate multiple factors is often hampered by the interventions that are aimed to optimise care. Although most models incorporate death and liver transplantation into one outcome, liver transplantation censors mortality, and some patients who have received a transplant would have survived if their clinical course had not been interrupted by liver transplantation. For example, the Kings College Hospital Criteria (validated in adults with acute liver failure) and the Liver Injury Unit score could not be validated when the outcomes of death and liver transplantation were separated.[24] Overall, the Kings College Hospital Criteria perform poorly in paediatric acute liver

Panel 1: Key considerations for liver transplantation in paediatric acute liver failure

Eligibility
- The cause of paediatric acute liver failure (PALF) suggests a poor prognosis and mortality without transplantation
- Probability of survival for 5 years post-transplantation is greater than 50%
- Patient is well enough to survive the transplantation and recovery

Influential factors for decision making
- PALF disease trajectory and degree of multiorgan involvement
- Patient and family perspective on post-transplantation quality of life
- Likelihood of immunosuppression adherence (less relevant for children with strong social support)

Reasons to avoid liver transplantation
- Formal contraindications, including uncontrolled systemic infection, multisystem mitochondrial disease, and extrahepatic malignancy (panel 2)
- Evidence of brain herniation or brain death
- Severely reduced post-transplantation quality of life (eg, ongoing severe multisystem disease)

Panel 2: Absolute and relative contraindications for liver transplantation in paediatric acute liver failure

Absolute contraindications
- Generalised extrahepatic malignancy (with the exception of hepatoblastoma with pulmonary metastasis)
- Uncontrolled systemic viral or fungal infection
- Severe multisystemic mitochondrial disease
- Alpers syndrome
- Niemann–Pick disease type C
- Severe refractory portopulmonary hypertension

Relative contraindications
- Cardiovascular compromise (ie, requiring high vasoactive support)
- Multiorgan involvement
- High-frequency oscillatory ventilation
- Haemophagocytic lymphohistiocytosis
- Valproate-induced liver failure associated with *POLG* mutations
- Substance use
- Severe obesity
- Severe acute respiratory distress syndrome with ratio of arterial oxygen partial pressure to fractional inspired oxygen below 100 mm Hg
- Poor pre-existing physiological reserve or frailty
- High graft-to-recipient weight ratio
- High likelihood of treatment nonadherence despite multidisciplinary interventions and support
- Social, psychological, or economic concerns

Adapted from Squires and colleagues.[12]

failure; in one study, the criteria had a 33% positive predictive value for death.[25] Modelling efforts in the past decade have used dynamic prediction strategies based on the trajectory of measurements over time (table),[21,22] but the inclusion of variables that can be subjective and difficult to assess (eg, encephalopathy) or modifiable by supportive measures such as blood product infusion (eg, prothrombin time/INR, fibrinogen), extracorporeal support (eg, ammonia, bilirubin, lactate), or supplementation (eg, phosphate, albumin) continues to limit their clinical effect.[16] Chronic liver disease scoring models, including the Pediatric End-Stage Liver Disease and the Model for End-Stage Liver Disease scores, are occasionally applied with variable success.[26] Artificial intelligence modelling has yet to be extensively explored; in one small study (n=54), an artificial neural network showed excellent discriminative ability (area under the receiver operating characteristic curve of 0·96),[27] but such approaches require validation in larger cohorts.

Intensive care considerations

Children with PALF are often admitted to intensive care units for monitoring and treatment of the three primary drivers of mortality: cerebral oedema, uncontrolled sepsis, and multiorgan failure.[28] These complications, when severe, can mean patients become ineligible for liver transplantation.

Firstly, children who develop encephalopathy as part of PALF are at increased risk of death from cerebral oedema and brain herniation.[29,30] Neuroprotective strategies, alongside formal neuromonitoring, are designed to prevent or treat cerebral oedema by maintaining normocarbia and normothermia and avoiding hypoxia, hyponatraemia, and stressors that could exacerbate intracranial pressure.[6] Non-invasive intracranial pressure monitoring is preferred over invasive monitoring because of the risk of bleeding associated with the latter.[31] Non-invasive modalities include electroencephalogram, transcranial Doppler, cerebral near-infrared spectroscopy, and monitoring of the optic nerve sheath diameter. Managing intracranial pressure is designed to avoid progressive intracranial hypertension, coning, brain death, and catastrophic intracranial bleeds or strokes, all of which would represent contraindications to liver transplantation.[32]

Secondly, uncontrolled sepsis is often a contraindication to liver transplantation. The risk of sepsis is high in PALF because the anti-inflammatory response to liver injury results in functional monocyte deactivation.[33] Use of broad-spectrum antimicrobials is therefore critical.[34]

Finally, PALF can result in multiorgan failure; 51% (45/88) of the Pediatric Acute Liver Failure Study Group cohort who died, died of multiorgan failure.[30]

	Location	Cohort size (n)	Aetiology	Variables	Timepoint	Outcome	AUROC
Dynamic Prediction Model,[22] 2024	USA, Canada, and UK	1144	Mixed: 45% indeterminate, 12% acetaminophen, and 10% metabolic	INR, total bilirubin, hepatic encephalopathy, creatinine, and platelet count	Dynamic: serially over 3 days	Mortality at 21 days	0·887
Children's Hospital of Los Angeles acute liver failure,[20] 2023	USA	147	Mixed: 35% indeterminate, 24% drug-induced, and 17% autoimmune hepatitis	Albumin, ammonia, and total bilirubin	Admission	Survival with native liver	0·78
PALF-Delta Score,[21] 2020	South Korea	146	Mixed: 49% indeterminate and 16% metabolic	Total bilirubin, INR, and ammonia	Dynamic: change in peak and daily total bilirubin, peak and daily INR, and peak ammonia	Survival with native liver	0·947
Peds-HAV,[16] 2020	India	120	Hepatitis A virus	Hepatic encephalopathy, INR, and jaundice-to-hepatic encephalopathy interval	Peak hepatic encephalopathy grade 3 or 4 and INR >3·1	Survival with native liver at 28 days	0·921
Liver Injury Unit Score,[15] 2006	USA	81	Mixed: 20% indeterminate, 20% sepsis or ischaemia, and 14% metabolic	Total bilirubin, prothrombin time or INR, and ammonia	Peak	Survival with native liver at 28 days	0·905
Queensland Liver Transplant Service,[27] 2013	Australia	54	Mixed: 31% indeterminate, 26% metabolic, and 15% infective	34 input variables	Varied: including on admission, diagnosis, day 7, and peak	Liver transplantation or death without transplant within 6 months	0·96

AUROC=area under the receiver operating characteristic curve. INR=international normalised ratio.

Table: **Prognostic models developed for paediatric acute liver failure (PALF)**

Initial liver injury can escalate into multiorgan injury through one of several mechanisms. For example, high cardiac output can be associated with distributive shock physiology. Severe hypoxia can occur secondary to acute respiratory distress syndrome (ARDS), fluid overload, pulmonary hypertension, and hepato-pulmonary syndrome. Multiorgan compromise is suggested by the presence of one or more of the following: high doses of inotropes or vasopressors, high ventilator pressures or FiO_2 requirements, severe acute kidney injury, elevated pulmonary artery pressures, high intracranial pressures, or low cerebral perfusion pressures.[35] Severe acute kidney injury is common,[36] and continuous renal replacement therapy (CRRT) is usually initiated for both renal and hepatic indications (to treat hyperammonaemia or hepatic encephalopathy).[37] For clinically unstable patients in multiorgan failure, bridging therapies such as plasma exchange can be considered to stabilise the patient until they might be suitable for liver transplantation reassessment. Plasma exchange provides a survival benefit in adults with acute liver failure[10] and is recommended by the American Society for Apheresis and the European Association for the Study of the Liver.[38,39]

As with the liver-specific prognostic tools, intensive care scoring models (eg, Pediatric Risk of Mortality and Sequential Organ Failure Assessment scores) have shown mixed results and might not be appropriate for PALF.[3,40] For critically ill adult patients with cirrhosis, the following thresholds of severity of organ failures that contraindicate liver transplantation have been set by international consensus: arterial oxygen partial pressure to fractional inspired oxygen ratio (PaO_2:FiO_2) less than 150 mm Hg, a norepinephrine dose greater than 1 µg/kg per min, and serum lactate above 9 mmol/L.[41] Frailty and uncontrolled sepsis (in the presence or absence of multiresistant bacteria) are reasons to delay liver transplantation.[41] No such consensus exists for PALF. With advancements in organ support, each patient is discussed by the multidisciplinary team. Extraordinarily, some patients with what is deemed reversible severe acute respiratory or cardiovascular failure have been bridged to liver transplantation after receiving perioperative extracorporeal membrane oxygenation (ECMO).[42–44] ECMO should only be used when pulmonary and cardiac function is anticipated to improve quickly.

For the intensivist, the first factors to consider when establishing suitability for liver transplantation together with the hepatologist are age and diagnosis. Intensivists

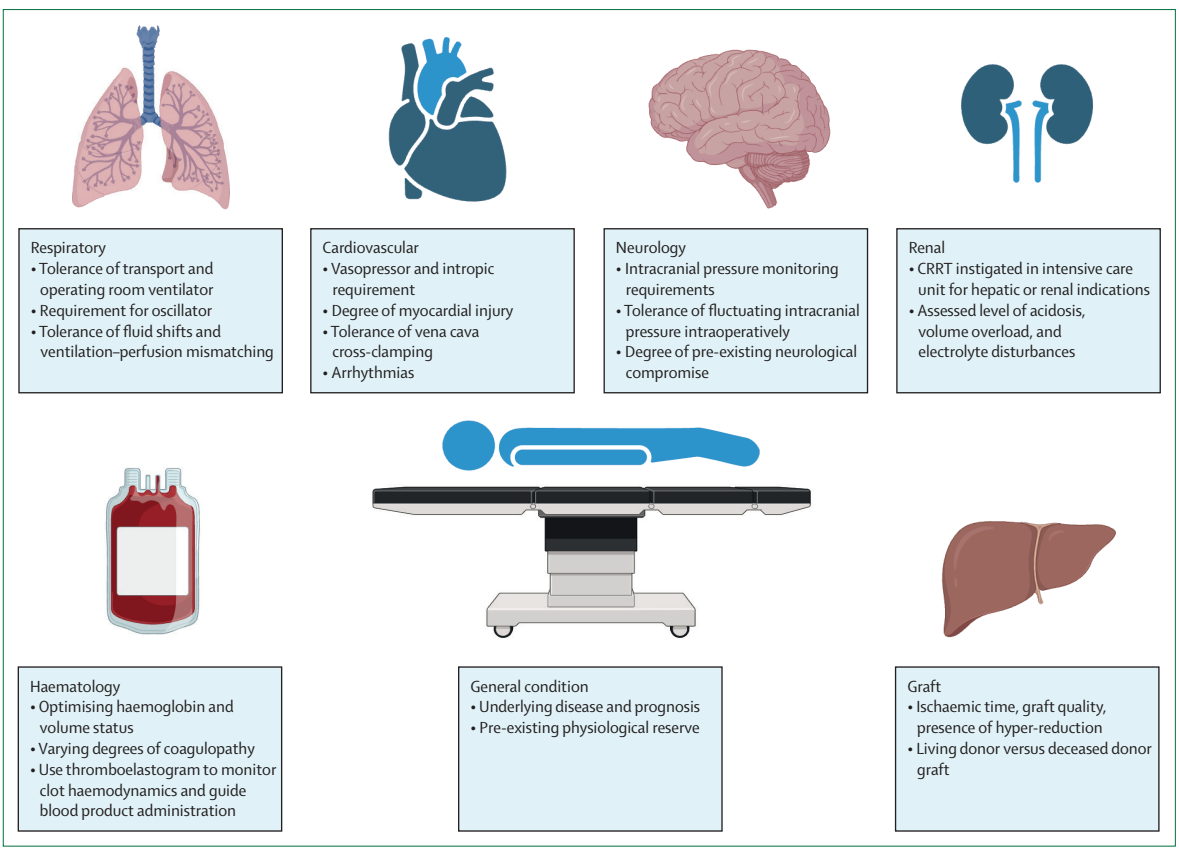

Figure 3: **Organ-specific perioperative considerations in relation to liver transplantation in paediatric acute liver failure**
Figure created with BioRender.com. CRRT=continuous renal replacement therapy.

must then evaluate reversibility and severity of end-organ failure. Patients with severe hepatic encephalopathy have a poor prognosis. The need for mechanical ventilation for severe ARDS, vasopressor support, CRRT, plasma exchange, and ECMO should not preclude liver transplantation in isolation but should be weighted alongside the overall probability of the patient surviving liver transplantation.

Anaesthesia and perioperative risks

Anaesthesia in PALF is high risk, primarily because of requirements for multiorgan support and low physiological reserve, particularly in neonates.[45] Anaesthetists must undertake a risk assessment to balance the insult of the surgical stress response, bleeding, and reperfusion injury against organ reserve and dysfunction (figure 3). Understanding the limits of local organ support techniques in the paediatric intensive care unit and operating room, including availability of CRRT or ECMO, informs whether liver transplantation is unsafe.

Organ-specific assessments of margin of safety

Respiratory

Disconnecting from the sophisticated intensive care ventilator during transfer of the patient to surgical theatre is the first hurdle of liver transplantation, and some respiratory reserve in oxygenation and ventilatory pressures is required. A degree of deterioration during transfer to theatre is acceptable; once the abdominal cavity is opened, ventilatory pressures generally improve, owing to improved pulmonary excursion. Rapid shifts in volume status result from intermittent caval compression during retrohepatic dissection, blood loss, and third spacing in pulmonary and intestinal vascular beds, which further complicate volume correction and heighten the risk of ventilation–perfusion mismatching and ventilatory difficulty intraoperatively.

Cardiovascular

Managing organ perfusion in the context of uncontrolled vasoactive mediator release in PALF requires a careful balance between volume resuscitation, vasopressors, and inotropes. Subclinical myocardial injury is common in PALF,[46] and the myocardium must be able to tolerate caval clamping and massive fluid shifts. Cross-clamping of the vena cava greatly reduces venous return, resulting in haemodynamic instability until caval flow is re-established. Arrhythmias are a worrying sign of toxicity, either due to direct effects on the myocardium or cerebral compromise.

Neurology

Liver handling intraoperatively releases vasoactive mediators that increase cerebral blood flow and intracranial pressure. Early tying of the hepatic artery can often provide haemodynamic stability and restore more normal levels of oxygen extraction. Intracranial pressure will drop when the liver has been devascularised, but surges again at the point of liver reperfusion.[47] Thiopentone and opiate infusions avoid the decoupling effect of volatile anaesthesia agents on the cerebral flow–metabolism relationship,[48] and obtund intracranial pressure surges when burst suppression is achieved (guided by bispectral index or continuous electroencephalogram).[49]

Renal

A child in established multiorgan failure will often have CRRT to aid in toxin removal (for neuroprotection), fluid and electrolyte balance, and haemodynamics.[50] Assessment of acidosis, volume overload, and electrolyte disturbances will guide decision making regarding intraoperative CRRT, depending on centre expertise and staffing. A child with renal failure will need potassium removal before reperfusion to avoid cardiac arrest and volume removal to accommodate transfusion and reduce the risk of pulmonary oedema.

Haematology

Optimising oxygen delivery requires management of anaemia and volume status. Blood loss and coagulopathy can be variable in PALF, with some reduction in both procoagulant and anticoagulant factors.[51] Conventional markers of coagulation do not provide an accurate estimate of the clotting profile;[52–54] however, the haemostatic system is shifted in the procoagulant direction, resulting in preserved clot stability.[52] Therefore, we recommend thromboelastography to monitor clot dynamics and guide decision making regarding volume or blood component replacement, mitigating risks of transfusion-related acute lung injury or circulatory overload, and transfusion-associated immunomodulation.

Graft selection

Graft reperfusion is a critical moment where patient stability is paramount. Reperfusion can be catastrophic if there has been prolonged graft ischaemic time and in specific graft conditions such as marginal grafts (deemed high-risk, typically on the basis of donor's demographic, clinical, laboratory, and histological data) and hyper-reduced grafts (optimised in size for the recipient child). Living donor grafts, with minimal ischaemic times and careful donor selection, could add a margin of safety but necessitate consideration of benefit versus risk to donor and family. Organ transplants from deceased donors have wide-ranging retrieval timings and variable graft quality. Upon graft reperfusion, vasoactive mediators, lactate, potassium, and cold fluid rapidly reach the right heart and pulmonary vasculature, inflicting intense pulmonary vasoconstriction and potentially right heart dilatation and failure. Graft reperfusion is therefore often associated with arrhythmias and hypotension.

Overall, it is common for children with PALF to become even more unstable during liver transplantation. Careful preoperative evaluation of the child's ability to withstand the procedure is crucial. A decision not to proceed on anaesthetic grounds is generally based on trends in overall organ dysfunction in the context of the graft quality.

If the child is on conventional ventilation, then maximal levels of positive end-expiratory pressure will make transfer very high risk, especially if the child already requires 70–100% oxygen. PaO_2:FiO_2 below 100–200 mm Hg is associated with high mortality in ARDS and can be another contraindication for transplantation.[35] Severe cardiogenic shock, maximal vasopressor and inotropic support, and evidence of severe right heart failure are prohibitive, and reassessment after improvement is advised. Irreversible brain damage is difficult to assess, but often these children have signs of profound haemodynamic instability and develop arrhythmias, therefore declaring them unsuitable.

Surgical considerations

Surgical management of PALF is profoundly challenging. The urgency of a life-saving operation hangs in a complex balance alongside the operative risks and potential benefits. Deciding to proceed with liver transplantation goes beyond technical feasibility and requires a synergistic approach from the multidisciplinary team.

A surgeon's initial concern is the patient's current clinical status and physiological reserve. Paediatric patients with no comorbidities will usually have a robust physiological reserve, whereas neonates have a comparatively limited physiological reserve. Children who are malnourished and, for example, develop PALF associated with hepatitis A virus infection are likely to have decreased physiological reserve. Reduced reserve is also relevant for children who develop PALF secondary to multisystem disorders such as mitochondrial disease. An assessment of physiological reserve informs the risk of perioperative and postoperative mortality; paediatric tools such as the PedsQL 4.0 Multidimensional Fatigue Scale can be used as part of this assessment.[55] Postoperative complications exacerbated by limited physiological reserve can be ameliorated through adequate nutritional support, but resource-poor facilities within low-income settings might be limited in what they can offer.

The transplant candidate's condition must be considered alongside the surgeon's assessment of the donor and graft quality. The donor's infection status (and the risk of transmission to the recipient), blood group compatibility, organ size, donor age, overall health, comorbidities, and the condition of the organ (including ischaemia time) are important elements of an evaluation

process that must be completed within a narrow window of time.[56]

Technical considerations and experience

A considerable technical challenge of liver transplantation in young and small patients relates to the graft-to-recipient weight ratio (GRWR); a large graft and high GRWR can cause considerable surgical complications, such as delayed abdominal and skin closure.[1,57] Patients with PALF also often present with preoperative and postoperative anasarca due to systemic inflammatory response syndrome or renal failure,[36] which increases the risk of postoperative infections in an already immunocompromised child.[58] A further technical concern is the requirement for oscillatory ventilation, which makes transfer and surgery difficult since the fine suturing of the anastomoses is made impossible.

Both surgeons and anaesthetists must be alert to the concomitant risks of PALF-associated bleeding and thrombosis that result from distorted synthesis of prothrombotic and antithrombotic factors.[59] This tendency to bleeding or thrombosis is even more challenging during liver transplantation due to the requirement for numerous vascular anastomoses, blood flow variability, and reperfusion.[60] This concern, again, is amplified in smaller patients and with high GRWR.[61,62]

These surgical challenges highlight the importance of a centre's collective experience and the surgeon's personal expertise. The correlation between surgeon experience and outcomes in technically demanding cases is well documented[63,64] and reinforces the notion that proficiency and deep understanding of the unique challenges of paediatric liver transplantation are crucial for achieving favourable outcomes.[65]

Social and psychological considerations

In most high-income countries, a child with PALF who requires a liver transplant will be able to access one. However, organ availability on a global level is strikingly varied. Not all medical systems can provide liver transplantation facilities or the advanced critical care required to manage these extremely unwell children. Poverty and the ability of the patient's family to pay for post-transplantation immunosuppressants are a real concern, with no clear solution. Evidence of financial strain exists even for transplant recipients in high-income countries.[66–68] Can a liver transplant be justified for a child whose family cannot afford immunosuppressants?

Social workers can be crucial facilitators, especially if they can arrange additional support for families facing financial strain.[68] However, social workers are not available in all countries, and health-care systems vary in their resources, not only with regards to social support provision but for all aspects of transplant-related care. The level of social support a child has (or does not have) could affect their suitability for transplantation, especially in health-care systems with limited resources.

Psychological considerations are equally important. Adherence to post-liver transplantation immunosuppressive regimens can be challenging, especially in adolescence.[69] In a hypothetical case, consider an adolescent who requires liver transplantation after an intentional drug overdose precipitating PALF. If the patient receives a liver transplant and subsequently presents with a further overdose and failure of the grafted liver, psychosocial evaluation of probable quality of life or recurrence of the same event will be extremely important when weighing up the risks and benefits of a second liver transplantation. Similar questions should be asked of patients who are not following treatment for pre-existing comorbidities. An example might be an adolescent with diabetes who develops acute liver failure[70] but has a medical history of multiple episodes of diabetic ketoacidosis due to non-adherence to insulin.

Ethical considerations: the moral choice

It is necessary to consider the ethical dimension of suitability for liver transplantation in PALF. Ethics is a fundamental part of difficult decision making in modern medicine and is therefore the business of all health-care professionals. The role of the ethicist is both to help with complex decision making, and support those who make such decisions. Formal ethical review requires resources and time, but a few key principles lend themselves well for consideration by all stakeholders of the PALF–liver transplantation journey.

Help with complex medical decision making

Traditional ethics distinguishes right from wrong and guides appropriate action, and most clinicians will be familiar with the four principles of clinical ethics: patient autonomy, beneficence, non-maleficence, and social justice.[71] However, these principles are of limited use in complex paediatric cases. Indeed, when considered alongside other ethical norms, such as deontology, utilitarianism, and virtue ethics, their application offers varied but conflicting solutions. Complex decisions in paediatric transplant medicine might have to incorporate further ethical norms, such as feminist ethics, and considerations of faith, culture, and world views of the child, their family, and their clinical team, as bounded by national laws.

In normal paediatric practice, medical treatment is usually guided by considering the child's best interest, a welfare approach that weights potential burdens against benefits. In transplant medicine, however, the desire to do good and maximise the wellbeing of an individual might conflict with the principle of justice if transplanting an organ to one patient deprives another patient who could also benefit from the transplant. This moral conflict arises in an ecosystem of limited organs. Allocating organs to the most unwell patients at high risk of death also risks organ waste, so a utilitarian perspective on organ allocation—aiming to maximise the wellbeing of the greatest number

of patients—persists. Transplant policies are therefore designed to target zero organ waste by recipient death, and no undue harm from the transplant if survival was likely without. Allocation of donor livers for transplantation must, therefore, balance the potential recipient's acuity of illness (overall need) with likelihood of survival.[72]

The rule of rescue in modern medicine (first coined by A R Jonson and later shaped by J McKie and J Richardson)[73] describes the imperative medical professionals feel to save the individual patient in front of them facing avoidable death without considering the opportunity cost of doing so. In transplant medicine, clinicians are influenced by the imperative to help the patient in front of them who is facing imminent death. The multidimensionality of the moral conflict emphasises the importance of shared decision making.

The uncomfortable reality is that the threshold at which a child is considered too unwell for transplantation depends on organ availability. In countries with very low supply of donor organs, liver transplantation policies might favour relatively healthy candidates who are likely to survive the procedure. In Spain, where normothermic regional perfusion in donation after circulatory death has almost cleared the transplant waiting list,[74] children with complex comorbidities might be listed more readily than in countries with reduced organ availability, where death of children on the waiting list, with or without neurodisability, chromosomal issues, or other comorbidities, is more common. In countries with limited supply, organ rationing decisions might consider the patient's capacity to benefit from transplant, which uncomfortably might include the ability to derive benefit from life in terms of developmental potential, future independent living, and employment. Limitations in resources and expertise, especially in low-income and middle-income countries, could also restrict capacity or access to transplantation.

The role of living related donors presents an additional ethical quandary in PALF. Living relatives often have very little time to consider the morbidity and (rarely) mortality risks of becoming a donor. Furthermore, access to living-related organ donors is often inequitable; poor health in potential donors can limit their eligibility to donate. In some cultures, understandably, mistrust of health-care services hinders willingness to donate. In fact, inequity in donation and transplantation outcomes associated with poverty are serious challenges worldwide, even in countries that offer universal health coverage.

In the UK, the Royal College of Paediatrics and Child Health has devised a practical framework for decision making to limit treatment in life-limiting and life-threatening conditions.[75] According to this framework, treatment might be ethically withheld or withdrawn in three circumstances: (1) when life is limited in quantity; (2) when life is limited in quality; or (3) when a competent child consents to palliative care with the support of their clinicians and parents. In the context of PALF and liver transplantation, the criterion of a life limited in quantity would apply to a child confirmed to be brain dead by appropriately applied professional criteria and to a child facing imminent death (with physiological deterioration occurring irrespective of treatment), or inevitable death not necessarily imminent, but is expected such that prolonging life by liver transplantation confers no overall benefit.

Support for clinical staff

Ethical advice for clinical staff who are deliberating challenging decisions is increasingly recognised as a crucial element of the decision-making process. Clinical ethics support can take many forms.[76] When decision making has stalled because of internal disagreement or when complex decision making is required, we advocate for the involvement of formal clinical ethics services that can provide individualised advice.[76] Such services increasingly involve members of clinical teams, legal teams, lay public, patient support groups, and faith groups.[77] The family's perspective and the child's voice should be heard whenever possible, even relayed by others, such as family mediators. These services provide clinical teams with an open and transparent review of their decision making and help ensure that social, legal, and moral components are part of the overall assessment of the child's best interest. Ethics services can also ensure that families feel heard. Importantly, the involvement of a clinical ethics service can manage the moral injury experienced by the medical team; for example, the ethics service can facilitate a post-case debrief to process emotional components of challenging situations.[78]

Ethical considerations are fundamental when evaluating which children are suitable, or not, for listing for transplantation—whether due to true physiological futility of liver transplantation, an overwhelming risk of not surviving long after liver transplantation, or—most challengingly—the need to prioritise other patients with better chances of benefitting from the liver transplant.

As described at the start of this Review, for children who do not receive a transplant, there are three pathways. The first is ongoing intensive care, which continues until the child is stable enough for transplant or the child dies.[10,79] The second option is a total hepatectomy to render the patient anhepatic.[80] Although advances in supportive, neuroprotective, and extracorporeal therapies now render hepatectomy rare, the procedure can be indicated for toxic liver syndrome, in which hepatic necrosis propagates massive release of proinflammatory cytokines that cause gross systemic inflammation, cardiac collapse, and cerebral oedema. Hepatectomy theoretically minimises the spread of the cytokines, improving cardiovascular function and reducing cerebral oedema, and can potentially stabilise a patient who is otherwise too unwell for transplantation, but would be eligible if clinical stability was achieved. Although case series have described high mortality for this approach (22 [69%] in 32 patients in

Search strategy and selection criteria

Specialists from the hepatology, intensive care, anaesthesiology, transplant surgery, and ethics specialties identified references for this Review through searches of Pubmed and Google Scholar, from database inception until April 2024, using search terms relevant for each subsection, such as: (paediatric acute liver failure OR PALF) AND (aetiology OR pathology OR pathogenesis OR prognosis OR transplant OR listing OR outcomes OR anaesthesia OR anaesthetic OR surgery OR renal replacement OR kidney replacement OR ECMO OR allocation OR social OR psychological OR ethic*), as appropriate. Additional references were identified by searching the reference lists of already included articles. Only papers published in English were reviewed. The final reference list was collated based on the papers deemed by the authors to be the most cited, high quality, accessible, and relevant.

one series),[11,80] hepatectomy can rarely be considered as a last resort in specific challenging cases.

The last option for children with PALF is palliative and end-of-life care. The core principles of palliative care, including symptom management and support for the child and their family, are familiar to all paediatric professionals.[81] Families of children with PALF should be made aware of the severity of the condition from the point of admission and, if available, palliative care consultations can be introduced alongside active management, as this can facilitate early consensus and recognition of goals of care.[81,82]

Conclusions

Huge advances have been made in the management of children with PALF. An increasing proportion of children are not transplanted because ongoing intensive care leads to a realistic likelihood of native liver survival, outweighing the risks of transplant and life-long immunosuppression. However, we have shown that there are many reasons why a critically ill child with PALF will be unsuitable or ineligible for transplant. Importantly, such assessments are best made together, incorporating all members of the multidisciplinary team.

Contributors

AD conceived of and led the Review. All authors contributed to the first draft equally, and ECA refined and led revisions. All authors read and approved the final version for submission.

Declaration of interests

AD is chair of Scientific Affairs of the European Society of Pediatric and Neonatal Intensive Care (ESPNIC) and chair of the Working Group for Liver Failure in ESPNIC. JB is Chair of the Paediatric group of the Ethical Legal and Psychosocial Aspects of Transplantation of the European Society for Organ Transplantation. JS declares royalties from Up To Date. All other authors declare no competing interests.

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